

Purpose: Pre-Processing and Association Analysis.

Problem: A dataset (named **Economy**) will be e-mailed to your student account which is made up of 7 independent variables and one dependent (aka Class) variable named **Treat**. Each independent attribute represents a macroeconomic outcome.

treat	debt	lopen	lrer	bndvol	gdpgvol	pivol	spreadvol
1	26.4	78.4	99.4	0.4	1.2	0.2	1.1
1	26.4	78.4	99.4	0.4	1.2	0.2	1.1
1	6.2	85.5	96.2	0.3	1.1	0.4	1
1	23.5	87.6	96.4	0.5	0.5	0.7	0.6
1	25.3	87	97.1	0.5	1	0.4	0.1
0	36.3	86.5	100.5	0.4	1	0.5	0.1
0	25.3	90.9	101.8	0.4	0.6	0.4	0.1
0	6.5	94.2	101.3	0.5	0.8	0.5	0.2

The owner of the dataset wants you to complete the following three parts for their given data: [When it comes to your grading, the three parts serve as three assignments)

PART 1: (Determine the outliers within each attribute, redundant attributes, and discretize the dataset)

1- To determine the outliers in a given attribute,

- Calculate the standard deviation (STD) for the attribute and the average (AVG). Mark any value in the attribute that is greater than $(AVG + STD * 2)$ or less than $(AVG - STD * 2)$ as an **outlier**.
- Report the name of the attributes which have an outlier value along with the outlier value itself.
- After marking the outliers for all attributes, then remove those records from the dataset **Economy** that contains at least one outlier value.
- Report the number of records that have been removed and the number of the remaining records in the dataset.

2- Identify redundant attributes using Pearson correlation coefficient. Two attributes with correlation coefficient of greater than 70% (for example) are highly correlated. Report a correlation matrix as follows:

	A	B	C
A			
B			
C			

Each empty cell shows the correlation between the two attributes that are used as the row and the column names. Remember that the correlation coefficients for attributes (A, B) and (B, A) are the same. Therefore, the upper and lower triangles of the matrix are the same. The user of your system must be given a chance to select the threshold for the Pierce Correlation. The threshold value must be printed along with the correlation matrix. If the Pierce correlation coefficient shows that attributes A and B are correlated, then remove one of them from the dataset Economy.

By now, your dataset Economy is free of outliers and correlated attributes.

- 3- Discretize every attribute in the dataset Economy delivered by Step 2. To do so, use Entropy-Based discretization. (The dependent variable, Treat,) has already been discretized into two values of 1 and 0.

Let us assume that you are discretizing attribute A, for example, shown in (Figure 1-a).

A	A	A	Diff
5	13	13	
12.5	12.5	12.5	0.5
11.5	11.5	11.5	1
8.6	9	9	2.5
7	8.6	8.6	0.4
6	7.8	7.8	0.8
5.9	7	7	0.8
1.5	6	6	1
9	5.9	5.9	0.1
7.8	5	5	0.9
2.1	2.1	2.1	2.9
13	1.5	1.5	0.6

(a)

(b)

(c)

Figure 1: An example for discretization: (a) Attribute A, (b) sorted Attribute

- Sort the values in attribute A and in descending order (Figure 1.b)
- Get the difference between two adjacent values of A in Figure 1.b. The results are shown in Figure 1.c.
- Sort the values in Diff column and get the three largest numbers in Diff. For our example, they are: **Large Diff={2.9, 2.5, 1}**.

- (iv) The difference of 2.9 appears with value of {2.1}.
 The difference of 2.5 appears with value of {9}, and
 The difference of 1 appears with values of {11.5, 6}.

$$\text{Diff_Union} = \{2.1\} \cup \{9\} \cup \{11.5, 6\} = \{2.1, 9, 11.5, 6\}$$

We used steps (i to iv) to build the Diff_Union. After that there is no need for keeping the sorted attribute A and we use the attribute A shown in Figure 1.a.

- (v) As a reminder,

Entropy Based Discretization:

Let attribute A be a part of dataset, S, and it needs to be discretized.

Let us also assume that there are k unique values (v1, ..., vk) in A.

Step 1- *Pick a value from A as a Threshold value, T1.*

Step 2- *Partition the dataset S into two partitions of S1 (includes all the records for which value of A < T1) and S2 (includes all the records for which value of A ≥ T1).*

Step 3- *Calculate information gain in reference to T1. The I(S, T) is calculated as follows:*

$$\frac{|S_1|}{|S|} \text{Ent}(S_1) + \frac{|S_2|}{|S|} \text{Ent}(S_2)$$

Step 4- *Repeat Steps 1 through 3 for every unique value in A and calculate the Information gain.*

At the end we have I(S, T1), . . . , I(S, Tk).

For our example the number of unique values is equal to 12. Since the number of unique values for attribute A may be large, then we limit ourselves to the values in the set **Diff-Union** set which is smaller.

PART 2: (Determine the valid Associate Rules)

1. Break the **discretized file of Economy** into two files. Make file Econ-TEST by randomly pick 10% of the records from the discretized Economy dataset (produced in PART 1) for which Treat= 1 and also randomly pick 10% of the records from the discretized Economy dataset for which Treat= 0. In addition, make file Econ-TRAIN from the rest of the records in the discretized Economy dataset. For this part of the assignment, we use only the Econ-TRAIN file. **The rest of the following steps are applied only on Econ-TRAIN dataset.**
2. Report each attribute name, discrete values for the attribute, and the range of values for which each discrete value stands. An example is given below. (Note: all values in this table are fake and no calculation is done to get discrete values and ranges.)

Attribute Name	Discrete Value	Range
A	1	[13-9)
A	2	[9-2.1)
A	3	[2.1- 1.5]
B	1	[0.023-2.45)
B	2	[2.45-3.92)
B	3	[3.92-8.43)
B	4	[8.43-11.32]
C	1	[4.3-6.2)
C	2	[6.2-7.1]

If an attribute in the given dataset originally appears with discrete values, then ranges are the same as the discrete values.

3. Change each unique discrete value for a given attribute (including the dependent variable) to a name that starts with the letter “I” followed by x, where x is an integer number starting from 1 and it will be incremented by one for the next unique discrete value. Repeat the process for each attribute and do not reset the x to 1. For example, the following dataset has three discretized attributes of A, B, and C. **We assume that attribute C is the dependent variable.**

A	B	C
1	2	1
2	3	1
3	4	2
2	1	1
2	1	1
1	3	2

The unique discrete values for A, are 1, 2, and 3 which are replaced by I1, I2, I3. The unique discrete values for B, are 1, 2, 3, and 4 which are replaced by I4, I5, I6 and I7. The unique discrete values for C, are 1 and 2 which are replaced by I8 and I9. As a result, the dataset changes into:

A	B	C
I1	I5	I8
I2	I6	I8
I3	I7	I9
I2	I4	I8
I2	I4	I8
I1	I6	I9

This is the same as saying we have 6 transactions as follows with transaction ids of 10 to 60.

T-ID	Items
10	I1, I5, I8
20	I2, I6, I8
30	I3, I7, I9
40	I2, I4, I8
50	I2, I4, I8
60	I1, I6, I9

Report the following Table (**Table A**):

Attribute Name	Discrete Value	Naming	Range
A	1	I1	[13-9)
A	2	I2	[9-2.1)
A	3	I3	[4.5-7.1]
B	1	I4	[2.1- 1.5]
B	2	I5	[2.45-3.925)
B	3	I6	[3.925-8.423)
B	4	I7	[8.423-11.32)
C	1	I8	[4.3-6.2)
C	2	I9	[6.2-7.1]

4. Apply the Apriori algorithm to identify the frequent itemsets and report them. An example is shown below:
 - (a) I5, I8, I2
 - (b) I1, I4, I9
 - (c) I6, I4, I3
 - (d) I8, I7, I4

In this example, the frequent itemsets are of length 3 (i.e., each frequent itemset includes three names). Keep in mind that Apriori algorithm in the first iteration creates valid frequent itemsets of length 1. In the next iteration, Apriori builds the frequent itemsets of length 2 out of the itemsets of length 1. In the next iteration, itemsets of length 3 are created out of the frequent itemsets of length 2. This process continues to create itemsets of length 4, 5, . . . The process stops when Apriori cannot build valid frequent itemsets.

5. Remove those frequent itemsets, from the above list in which, there are more than one name that belongs to the same attribute. For example, I4 and I7 appear in the frequent itemset (d) and both belong to the attribute B. Therefore, the frequent itemset (d) is dismissed.

If all the frequent itemset generated in the above step are removed, then use the valid frequent itemsets that their length is one less and repeat step 5.

Report the removed itemsets and the reason for their removal. (e.g., The reason for removal of (d) is: *more than one value of attribute B found in this frequent itemset*).

6. Generate all the association rules as we discussed in class for each frequent itemset.

For example, rules generated for the itemset I5, I8, I2 are:

$$\begin{array}{lll} I5 \Rightarrow I8, I2 & I8 \Rightarrow I5, I2 & I2 \Rightarrow I5, I8 \\ I5, I8 \Rightarrow I2 & I5, I2 \Rightarrow I8 & I8, I2 \Rightarrow I5 \end{array}$$

- a- Determine whether the dataset is *balanced* or *imbalanced*.

Dataset is imbalanced if the imbalance ratio (IR > 0.4).

- b- If the dataset is balanced, then

- Calculate and report the confidence for each association rule along with the rule itself.
- Report the list of survived association rules after applying the threshold confidence of 70%. The user of your system must be given a chance to select the threshold for confidence. The threshold value must be printed along with the list of survived association rules.

- c- If the dataset is imbalanced, then

- Calculate and report the Kulc confidence for each association rule along with the rule itself.
- Report the list of survived association rules after applying the threshold confidence of 70%. The user of your system must be given a chance to select the threshold for confidence. The threshold value must be printed along with the list of survived association rules.

7. Convert each rule delivered by the above step (Step 6), into plain English using the following two formats separately (Format-1 and Format-2).

Format-1: Let us assume that the following association rule is one of the final association rules:

$$I9, I5 \Rightarrow I2 \quad (\text{confidence} = 80\%)$$

The conversion of this rule in format 1 is :

$$C = 2 \wedge B = 2 \Rightarrow A = 2 \quad (\text{confidence} = 80\%)$$

We use information from **Table A** for this conversion.

You report the list of all the converted rules in format-1. (We refer to this list as the **Format-1- List**).

Format-2: Let us use the same rule.

$$I9, I5 \Rightarrow I2 \quad (\text{confidence} = 80\%)$$

The conversion of the rule in format-2 is :

$$C [6.2-7.1] \wedge B [2.45-3.95] \Rightarrow A [9-2.1] \quad (\text{confidence} = 80\%)$$

We also use information from **Table A** for this conversion.

You report the list of converted rules in format 2. (We refer to this list as the **Format-2- List**)

PART 3: (Prediction of the dependent variable using the association rule produced in PART 2.)

1. Remove from the **Format-1 List** all the association rules that (a) their conclusions (i.e., the right side of the implied sign) have more than one predicate and (b) the conclusion is not the attribute **Treat**. The resulting list may be a smaller version of the Format-1-List and name this version **Format-1-Rules**.
2. Use these rules to predict the dependent variable for the records in Econ-TEST. How?

Suppose one of the rules in Format-1-Rules is $C = 2 \wedge B = 2 \Rightarrow A = 2$. The rule has a set of conditions (the left side of the implied sign) and a conclusion (the right side of the implied sign). Check all the records of the Econ-Test dataset one by one for the conditions. If you find a record that has the value of 2 for attribute C and value of 2 for attribute B, then you predict that the value for the dependent variable (A) is 2. If the record has the value of 2 then your prediction is correct (i.e., true), otherwise your prediction is false.

You need to count how many correct and incorrect predictions you have for every unique value of the dependent variable. If our dependent variable has only two values of 1 and 0, then you must create and report the following matrix:

Unique Values in the Dependent variable (Decision attribute)	Predicted Values	
	1	0
1	X	Y
0	Z	W

X is the number of records in Econ-TEST that their actual values for the dependent variable are 1 and their **predicted values** are 1 too (I.e., the prediction was correct). X is known as true positive (TP).

Y is the number of records in Econ-TEST that their actual values for the dependent variable are 1 and their predicted values are 0 (I.e., the prediction was incorrect). Y is known as false negative (FN).

Z is the number of records in Econ-TEST that their actual values for the dependent variable are 0 and their predicted values are 1 (incorrect prediction). Z is known as false positive (FP)

W is the number of records in Econ-TEST that their actual values for the dependent variable are 0 and their predicted values are 0 too (correct prediction). W is known as true negative (TN).

Report the calculation of the following indices:

Accuracy = $(X+W)/(X + Y + Z + W)$,

Recall = $X/(X+Y)$, and

Precision = $X/(X+Z)$